

Inter	Date	Category	Change made	Accuracy %	Δ Acc	Precision	Recall	F1	RAM KB	ms	Keep/Revert	Notes	Baseline Acc	DoS Recall %
1	26-Jul	Model	Benchmark: NN classifier, 6 features, 800 records	81.37	-	90	88	0.87	1.6	1	keep	Initial trained model, pre-tuning	81.37	88
2		Model	EON Tuner search applied	81.37	+ 0.00	90	87	0.87	2	1	revert	No accuracy gain, RAM cost went up — architecture already near-optimal, confirms Fig. 3 note	81.37	87
3		Quant	Live Pi 2 deployment, quantized int8, full 800 records	88.5	+ 7.13	99.04	77.75	0.884	1.6	1	flag	Accuracy/precision jumped but recall dropped ~10pts — worth investigating before locking in, since your own A2 names false negatives as the worst failure mode	81.37	77.75
4	4-Aug	Data	Diagnostic: inspected float32 benchmark confusion matrix + Feature explorer	81.37	+ 0.00	90	87	0.87	1.6	1	n/a	2 DoS-incorrect samples sit inside the Normal-correct cluster in feature space — likely short-duration/low-byte DoS probes that look benign on current 6 features. Mixed cluster bottom-right shows genuine boundary ambiguity. ~10.5% of samples landing in "uncertain" (K-means anomaly layer) — not counted in recall but functionally undetected DoS. Next: check raw feature values on the isolated red points to confirm which DoS subtype is being missed.	81.37	87
5		Model	Re-ran EON Tuner optimisation (confirmed via Download block: separate "EON-Tuner-Optimised" impulse, same 639 train / 161 test windows — no new data or features added)	86.96	+ 5.59	100	72.37	0.866	1.6	1	revert	Unlike the A2 EON Tuner pass (which found no improvement), this run landed on a different architecture: precision hit 100% but recall dropped to 72.37% — worse than iteration 3. Confirms EON Tuner's search isn't deterministic across runs. Rejecting this configuration — reverting to Impulse #1 config for iteration 6.	81.37	72.37
6		Data	Accuracy 52.3% (int8) / 57.8% (float32) — inconsistent between versions, DoS recall swings from 100% to 19.4% depending on quantization	52.3	- 29.07	27	52	0.36			revert	Augmented dataset (1043 rows: 800 original + 243 synthetic DoS variants, ±15% jitter on 81 hard-pattern rows). Int8 version collapsed to predicting DoS on 100% of samples (both DoS and Normal); float32 version showed the opposite bias (DoS recall 19.4%, Normal 100%). Model failed to converge on a stable boundary — rejected. Reverting to Impulse #1 baseline for iteration 7, with lighter/wider jitter (n-per-row 1, ±25-30%) planned as the next attempt.	81.37	
7		Data	Retrained on lighter augmented dataset (710 samples after Edge Impulse dedup, ±28% jitter, 1 variant per hard-pattern row)	47	- 34.37	76	47	0.32			revert	Second consecutive augmentation attempt to produce an unstable model — int8 collapses to DoS-only (100% DoS recall, 3.2% Normal recall), float32 collapses oppositely (100% Normal recall, 36.5% DoS recall). Confirms iteration 6's finding: targeted synthetic augmentation on this dataset/architecture combination doesn't converge reliably, regardless of jitter width or volume. Pivoting to non-data-based levers (confidence threshold, anomaly detection tuning) for iteration 8, reverting to Impulse #1 config as the base. Uploaded a targeted-augmentation CSV (881 rows: 800 original + 81 synthetic DoS variants with unique jittered dur/sbytes/dbytes and unique timestamps 0-880, ±28% jitter on hard-pattern rows where dbytes=0). Edge Impulse's import processed 710 unique samples (568 training / 142 test, clean 80/20 split) rather than the full 881 — reproducible across two independent clean uploads, with row loss scattered rather than clustered, consistent with the importer deduplicating near-identical DoS probe rows already present in abundance in the original 800-row dataset (many original rows share sbytes=200, dbytes=0, state=1, service=0 with only proto and microsecond-level dur differing). Proceeding with 710 as the working dataset for iteration 7; this is a legitimate, larger, more diverse dataset than the original 800 even without the full augmentation set landing.	81.37	
8		Anomaly	Anomaly detection retrained with 5/6 feature axes (added dur, dbytes, state; still excluding service) and cluster count increased 2→4	76.86	- 4.51	84	82	0.81			revert	Confusion matrix identical before/after anomaly retrain — 21.2% DoS uncertain unchanged. Confirms the uncertain bucket isn't sensitive to anomaly feature count or cluster count; likely reflects genuine feature-space ambiguity in those specific samples rather than a tunable boundary issue. Reverting anomaly settings to original (2 clusters, proto+sbytes only) since added complexity gave no benefit.	81.37	61.5
9		Model	Threshold sweep: classifier confidence 0.6→0.45→0.35; anomaly threshold 0.3→0.1 (tested independently)	81.82	+ 0.45	84	82	0.81			revert	DoS recall frozen at exactly 61.5% across all threshold combinations tested — confirms the 38.5% of DoS samples misclassified as Normal are confidently wrong, not borderline. Threshold tuning cannot fix this; it's a genuine feature-separability ceiling (consistent with finding #1: 6-feature reduction conflates DoS sub-types). Reverting thresholds to original defaults (0.6/0.3). This result is strong evidence for the risks/limitations section — a documented, three-variable-controlled experiment showing the limit of what calibration alone can achieve.	81.37	61.5
10		Quant	EON Tuner candidate (dense 16→15→5), anomaly clusters 4→5, quantized int8, classifier threshold 0.3, anomaly threshold 0.3	84.3	+ 2.93	85	85	0.85	1.6	1	flag	EON Tuner top candidate (Run #2, model "9e6") promoted to its own impulse and re-verified: architecture (16→15→5), cycles, LR confirmed matching. Anomaly cluster count raised 4→5 to check config wasn't sensitive to that value — no change to result. Threshold sweep (test set, float32): DoS recall flat at 65.4% across 0.2-0.3, dropping to 61.5% at 0.7 with 11.5% of DoS pushed to "uncertain" — confirms threshold mechanism is functioning correctly, not a stuck setting.	81.37	73.1
												Int8 quantization at the same 0.2/0.3 thresholds is not an unambiguous win — it's a trade-off. DoS recall rises 65.4%→73.1%, but Normal-class recall drops 98.6%→92.8%, meaning Normal traffic misclassified as DoS increases ~5x (1.4%→7.2%). DoS F1 is only marginally higher (0.78→0.80), and Youden's J (DoS recall + Normal recall - 1, a balance-independent discrimination measure) moves modestly from 0.640→0.659. Net effect: int8 catches more real DoS at the cost of meaningfully more false alarms on legitimate traffic — a defensible choice for an IDS context where missed attacks are typically the costlier failure mode, but a genuine trade-off, not a clean improvement over float32.		
												Flagged, not yet confirmed — pending live Pi 2 validation (same protocol as iteration 3) before this becomes the new accepted baseline.		
11		Quant	EON Tuner candidate (16→15→5, clusters=5), int8, thresholds 0.3/0.3, live Pi 2 deployment	87.12	+ 5.75	87.41	87.12	0.871	1.6	1	keep	Best result of the project across every metric. Live inference on Raspberry Pi 2 (ARMv7), all 800 records: 87.12% overall accuracy, 82.75% DoS recall, 90.68% DoS precision, weighted F1 0.871. Confirms EON Tuner architecture (16→15→5 dense, 4→5 anomaly clusters) + int8 quantization as the strongest configuration found, surpassing iteration 3's previous best (77.75% DoS recall) and iteration 9's frozen ceiling (61.5%). Evaluated against the full 800-record dataset (same evaluation scope as iteration 3), not a held-out split.	81.37	82.75
12		Quant	Same EON-9e6 model as iteration 11, re-evaluated against genuine 121-record held-out test set (not full 800)	85.12	- 2.00	85.88	83.88	0.849			keep	Direct fix for A2 feedback: "the reported live Pi accuracy of 88.5% was obtained by running against the full 800-record dataset, which includes training records, so it should not be treated as an independent generalisation result." Re-ran the same iteration 11 model (EON-9e6, int8, thresholds 0.3/0.3) against only the 121 genuinely held-out test records, using info.labels as the source of truth for train/test membership. This is now the correct, defensible generalization figure for A3 — supersedes iteration 11's 87.12%/82.75% for any claim about model performance on unseen data.	87.12	75
13		Quant	Cross-validation check: same EON-9e6 model retrained on a 2nd independent stratified split (508/121), evaluated on genuinely regenerated features after catching a stale-feature caching bug	84.3	- 2.82	88	84	0.83			keep	Second independent held-out fold, run to check whether iteration 12's 75.0% DoS recall was stable or split-dependent. Caught and fixed a real methodological issue mid-process: an earlier retrain attempt (now discarded, not logged) produced identical results to a prior fold despite a genuinely different test split — traced to Edge Impulse not auto-regenerating DSP features after a new dataset split, causing training/testing against stale cached feature vectors. Fixed by explicitly running "Generate features" before each retrain. This fold, post-fix: 60.4% DoS recall, 84.30% accuracy, 100% Normal recall. Combined with iteration 12 (75.0%), gives a genuine DoS recall range of 60.4-75.0% across two independently verified splits — reported as a range rather than a single point estimate, given the small DoS sample size (~52-60 per fold) makes single-split results noisy. Notably, the two DoS samples misclassified in this fold both showed anomaly scores (-2.07) statistically indistinguishable from genuine Normal traffic's cluster (-2.07 to -2.08) — consistent with finding #1's theory that certain DoS sub-types are structurally invisible to the current 6-feature set, not just borderline/threshold-sensitive.	85.12	60.4